The abundance plot

The dynamic range of the experiment, drawn as every feature in rank order, and where any one of them sits in it.

AT A GLANCE

Where	Quality control → Abundance
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The normalized, unfiltered, non-imputed QC matrix, converted to linear values before averaging; after VSN, the unnormalized data.
One point	One feature in one condition: its mean intensity over that condition's replicates, on a $\log_{10}$ axis.
Two renderers	**Type** Normal draws a static figure; Interactive opens a hoverable plot in a new tab.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

A PROCESSED ABUNDANCE SCALE

Normalization changes the values ranked here. VSN is the exception: its output is only log-like, and exponentiating it does not give a linear intensity, so after VSN normalization this page uses the unnormalized data of the same rows, and a note under the plot says so: VSN output is not on a linear intensity scale, so this plot uses the

unnormalized data. Record the normalization and treat this as a relative abundance overview, not an absolute concentration measurement.

1 What the plot answers

The abundance plot shows how far the experiment reaches. Sorting every feature by its intensity and plotting it against its own rank turns the dataset into one curve, and the shape of that curve says how many orders of magnitude were covered, how quickly the signal falls away, and where the measurement runs into the floor.

It also places a selected feature within the measured intensity range. A feature near the right-hand end has a relatively low recorded intensity. That does not establish an instrument detection limit or an absolute concentration: peptide response, ionisation and data processing also affect intensity. Use missingness and replicate measurements to assess how well that feature was quantified.

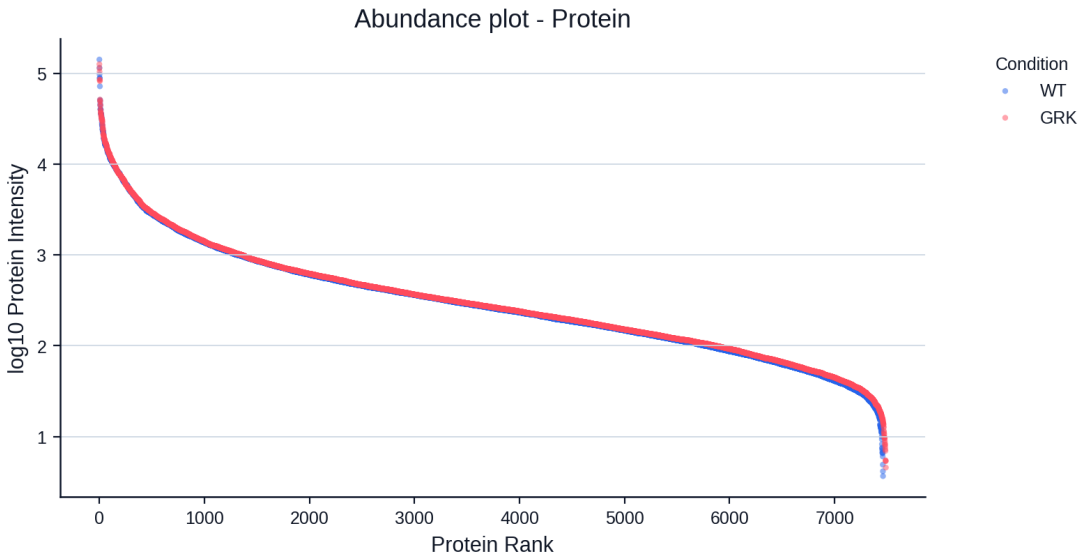


Figure 1.1. Protein level on the example dataset, All Conditions: the WT and the GRK curve drawn over one another, each running from rank 1 to about 7 500 and falling from above 10^5 to below 10^1 — four and a half orders of magnitude, a short steep shoulder, a long body and a tail that turns down at the right-hand end. The two curves are nearly indistinguishable, which is what two conditions of equal depth look like.

Reading it

- One point, one feature in one condition. Its height is the $\log_{10}$ of the mean of that feature over the condition's replicates.
- The x axis: the feature's rank within its own condition, 1 being the most abundant. It is an ordering, not a quantity — the distance between rank 10 and rank 20 means nothing beyond "ten features".

- The y axis: $\log_{10}$ intensity, so one unit is a factor of ten. At phosphoprotein level collapsed with robust_site_median it reads log10 Ratio to Site Median instead, because those values are ratios, not abundances.
- The colour: the condition. With one condition chosen there is a single curve; with All Conditions the curves are drawn over one another, and where they separate is where the conditions differ in depth.

EASY TO GET WRONG

The rank is computed separately for each condition, so a vertical line through the figure does not pick out one feature. Rank 500 in one condition and rank 500 in another are different proteins, and reading a gap between two curves at the same x as a fold change is wrong. To compare one feature between conditions, use the single-protein pages; to compare depth, compare the *lengths* of the curves and the height at which each ends.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Normal draws a static figure that can be downloaded. Interactive builds a Plotly figure with zoom and per-feature hover — name, condition, rank and intensity — opened in a new tab; there is no image download in that mode.	Normal	page
Condition	All Conditions draws every condition, each ranked within itself. Choosing one draws that condition alone, ranked 1 to *n*. The list is the conditions in the annotation, and a saved choice that no longer exists falls back to All Conditions.	All Conditions	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure, shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 10 cm, 300	page

Controls on this page (continued)

Control	What it does	Default	Where
Download format	PNG, JPG, WEBP, SVG or PDF. Normal mode only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Diameter in points, shape, opacity and outline — for all points at once, or for one condition at a time. The plot's own default is deliberately small: a dataset of eight thousand features drawn with larger marks is a solid band.	2.3 static All Conditions, 2.7 static single condition, 6 interactive	appearance
Condition names, Condition colors	One display name and one colour per condition, shared with every other plot in the session.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written from the data level: Abundance plot - Protein and its equivalents.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance
Frame and grid	The two renderers start from different conventions: spines on and no grid for the static figure, grid on and no spines for the interactive one.	by renderer	appearance

> **WORTH KNOWING**
>
> The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

1. Take the normalized, unfiltered, non-imputed QC matrix — after VSN, the unnormalized values of the same rows — and convert its stored values to linear values. Missing observations remain missing; they are not replaced by zero before averaging.
2. For every feature i and condition c, the arithmetic mean over the replicates in which it was quantified:

$$m_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\}.$$

3. The height of the point is $\log_{10} m_{i,c}$, with no pseudocount. The axis names the quantity, so the number has to be the logarithm of the quantity: adding one before taking the logarithm would change low values disproportionately: a mean of 1 would move from 0 to about 0.301, while a mean of 0.1 would move from -1 to about 0.041. Means that are not positive have no logarithm and are dropped rather than drawn at a value they do not have.
4. Within each condition the remaining features are ranked by $\log_{10} m_{i,c}$, descending, and the rank is the x coordinate. Ties are broken by the order the features appear in the matrix, so equal intensities take consecutive ranks rather than a shared average one.

WITH ONE CONDITION CHOSEN

The arithmetic is identical; only that condition's samples are averaged, and the ranks run 1 to n over the features that condition quantified. The static figure draws the same points in one colour; the interactive one adds hover text carrying the feature name, the condition, the rank and the intensity to four decimals.

Compare curve lengths and shapes, then inspect the features behind a difference. A shorter curve contains fewer quantified features; a vertical shift can reflect sample loading, normalization, measurement response or biology. Overlapping curves do not imply identical protein regulation, because each condition is ranked independently. Do not force curves to overlap merely to make the figure look consistent.

Edge cases

- A feature missing in every replicate of a condition has no mean there and is left out of that condition's curve. Curves therefore differ in length, and that difference is the point of the figure rather than a defect in it.

- A feature missing in some replicates is averaged over the ones it has, so a point can rest on fewer measurements than its neighbours.
- Features quantified nowhere at all are dropped from the figure entirely; the coverage table counts them under Identified, not quantified.
- With a single replicate in a condition the mean is that sample's own value, and the curve is that run's dynamic range.
- Interactive produces an HTML plot rather than an image: it opens in its own tab, and the download format selector is replaced by a link to it.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = abundance_plot(org_data, meta, workflow="Protein", data_scale="linear")
```

R

```
fig <- abundance_plot(org_data, meta, workflow = "Protein", data_scale = "linear")
```

workflow names the level — "Protein", "Phosphosite" or "Phosphoprotein" — and decides which column identifies a feature; data_scale states the scale of the frame handed in rather than converting it, and a $\log_2$ frame with data_scale="log2" works just as well — the function inverts the transform itself. There is no **Condition** argument: abundance_plot always draws every condition, and the single-condition curve is interactive_abundance_plot, which takes the condition as its third argument.

5 Exporting

In Normal mode the download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. In Interactive mode there is no image to download: the plot opens in its own tab, where Plotly's camera button saves what is on screen at screen resolution. Choose PDF for a figure that will be scaled or edited — at this many points a PDF is large but stays sharp — and PNG for a slide.

Behind this page

Function	What it produces
abundance_plot	The static figure: every condition, or one, ranked within itself
interactive_abundance_plot	The hoverable one, for a single condition. The package function additionally takes a list of features to highlight in red, and a limit on how many modification sites a phosphoprotein's hover label spells out; the page exposes neither
abundance_axis_label	Which y-axis label the level and the collapse method call for